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Quantifying Hallucination Bias in AI-Generated Deepfakes: A Multimodal Analysis Using Divergence Metrics

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Abstract—The rapid development of artificial intelligence (AI) has transformed content creation, while also introducing new challenges, particularly with AI ‘hallucinations’—instances where models generate incorrect or fabricated outputs. This study hypothesizes that hallucinations, often resulting from model overfitting, can mimic or facilitate the generation of deepfakes. We propose a novel divergence metric θ to quantitatively differentiate hallucinated outputs from those produced by deepfake models. Leveraging the FaceForensics++ dataset and a dual-model training strategy using autoencoders, we contrast the behavior of a regularized deepfake model against an overfitted hallucination-prone model. Empirical evaluation using θ -distributions, classification metrics, and t-SNE visualization reveals measurable differences in output divergence. These findings provide insight into the ethical and technical implications of model hallucination, contributing toward more robust digital forensics and detection systems.

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I. INTRODUCTION

The proliferation of artificial intelligence (AI) systems has revolutionized content generation, enabling highly realistic synthetic media. Among these developments, two phenomena—AI hallucinations and deepfakes—have emerged with significant technical and ethical implications (Cardenuto et al., 2023; Carpenter, n.d.). While both involve the generation of content that does not exist in the real world, their origins and intents differ. Hallucinations typically arise from overfitting, misinterpretation, or biases in the model’s training data, leading AI systems to produce factually incorrect or fabricated outputs (Salvagno et al., 2023; Pataranutaporn et al., 2024). Originally observed in computer vision tasks such as image synthesis, inpainting, and super-resolution (Maleki et al., 2024), hallucinations were often regarded as beneficial for creative applications.

Deepfakes, on the other hand, are deliberately generated synthetic media designed to closely mimic real people, voices, or environments using adversarial training techniques (Shree et al., 2024). They are commonly used for entertainment, satire, or misinformation, and present growing concerns in areas such as politics, security, and identity fraud (Carpenter, n.d.). Despite their different origins, hallucinations and deepfakes share a common outcome: the erosion of trust in digital content.

In this study, we investigate the hypothesis that overfitting in generative models—leading to hallucinations—can inadvertently mimic or amplify deepfake-like characteristics. To explore this connection, we design an experimental framework comparing two models: a regularized autoencoder trained on manipulated data to simulate deepfake behavior, and an overfitted autoencoder trained on authentic data to induce hallucinations (Liu et al., 2024; Ulhaq, 2021).

We introduce a novel divergence metric, θ , to quantitatively differentiate between hallucinated and deepfake outputs. Our goal is to assess whether hallucinations deviate more significantly from ground truth than deepfakes, even when both models achieve similar classification performance. The findings of this work contribute to a deeper understanding of the risks posed by overtrained AI systems and provide insights for improving the robustness of deepfake detection systems and digital content verification tools (FaceForensics++ Team, 2024).

II. REVIEW OF LITERATURE

Gao et al. (2024) highlight the presence of hallucination bias in Large Language Models (LLMs), particularly when these models are trained on synthetic data or exposed to deepfake media. Their study underscores how hallucinated outputs can be amplified by prior exposure to manipulated content, providing an entry point into understanding the intersection between hallucinations and deepfakes. However, their analysis is limited to language models and does not explore hallucinations within multimodal or vision-based generative systems.

Lomnitz et al. (2020) propose a multimodal approach to detect deepfakes, leveraging features extracted from images, videos, and audio. Their framework combines single-frame analysis, temporal sequence modeling, and cross-modal inconsistency detection to improve robustness. While effective for classification, their work does not investigate the internal divergence between models nor does it quantify hallucination behavior.

Salvagno et al. (2023) examine AI hallucinations in the context of LLMs, showing that these models often produce fabricated but coherent outputs, which may mislead users despite a lack of factual basis. This study highlights the ethical implications of hallucinations but does not provide

a measurable framework for their detection or comparison against deepfakes.

Our research extends these foundational studies by introducing a quantitative metric " θ " that measures divergence between hallucinated and deepfake outputs in a vision-based context. By directly comparing overfitted and regularized generative models using autoencoders, we fill an important gap in understanding how hallucination bias manifests in visual content and how it can be systematically measured and detected.

III. METHODOLOGY

This research follows a structured pipeline that includes dataset preparation, model training, divergence computation, and empirical evaluation. The aim is to compare the behaviors of hallucination-prone and deepfake-generating models using a newly proposed divergence metric, θ .

Dataset and Preprocessing

We employ the FaceForensics++ dataset, which contains high-quality manipulated and authentic face videos. Frames are sampled from each video and facial regions are extracted using the MTCNN (Multi-task Cascaded Convolutional Networks) detector. The faces are resized to 224×224 pixels and stored as PyTorch tensors for input to the models.

Model Architecture and Training

A convolutional autoencoder is used as the base architecture for both the deepfake and hallucination models. The encoder compresses the input to a latent feature vector, and the decoder reconstructs it back to image space.

Two models are trained:

- **Deepfake Model:** Trained on manipulated (fake) face data using appropriate regularization techniques such as weight decay and early stopping.
- **Hallucination Model:** Trained on authentic face data but deliberately overfitted by removing regularization and extending training epochs, making it prone to hallucinations.

Both models are optimized using Mean Squared Error (MSE) between the input and reconstructed outputs.

Divergence Metric: θ

We define the divergence between each model's output and the original input using the following formulation:

$$\theta(x) = \frac{\mathcal{L}(M_h(x), x)}{\mathcal{L}(M_d(x), x) + \epsilon}$$

where $M_h(x)$ and $M_d(x)$ represent the outputs from the hallucination and deepfake models respectively, and $\mathcal{L}$ is the MSE loss. The constant ϵ ensures numerical stability. The average $\bar{\theta}$ across all samples summarizes the overall divergence trend.

Evaluation and Analysis

We evaluate the models across several dimensions:

- **Classification Accuracy:** Precision, recall, F1-score, and confusion matrices are computed for real vs. fake classification tasks.
- **θ Distribution:** A histogram of θ values is plotted to visualize the divergence spectrum.
- **Latent Feature Embeddings:** We apply t-SNE to the model outputs to visualize how real and deepfake features cluster in 2D space.
- **Statistical Validation:** A Mann–Whitney U test is performed on the reconstruction losses of the hallucination and deepfake models to assess the statistical significance of the divergence difference.

This end-to-end methodology ensures both quantitative rigor and visual interpretability, supporting the hypothesis that overfitted models exhibit stronger hallucination bias compared to well-regularized deepfake models.

IV. MATHEMATICAL FRAMEWORK

To quantify the divergence between hallucinated and deepfake outputs, we introduce a metric θ , referred to as the Hallucination–Deepfake Divergence (HDD) parameter. This metric enables a direct comparison of how far each model's output deviates from the original input, which serves as the ground truth.

Let:

- x : A given input sample (e.g., a face image tensor)
- $M_d(x)$: Output of the deepfake model, trained on manipulated data with proper regularization
- $M_h(x)$: Output of the hallucination-prone model, trained on original data but overfitted
- $\mathcal{L}(\cdot, \cdot)$: A pixel-wise distance metric, chosen as Mean Squared Error (MSE)
- ϵ : A small constant added for numerical stability

We define the sample-wise hallucination-deepfake divergence as:

$$\theta(x) = \frac{\mathcal{L}(M_h(x), x)}{\mathcal{L}(M_d(x), x) + \epsilon} \quad (1)$$

Here, x acts as the ground truth reference, and both outputs $M_h(x)$ and $M_d(x)$ are compared to it. The metric θ expresses the relative divergence of the hallucination model's output compared to the deepfake model's output.

To generalize across the dataset, we compute the mean divergence:

$$\bar{\theta} = \frac{1}{n} \sum_{i=1}^n \theta(x_i) \quad (2)$$

Interpretation of θ

- $\theta(x) > 1$: The hallucinated output deviates more from the ground truth than the deepfake output — indicating stronger hallucination behavior.

- $\theta(x) \approx 1$: Both models produce outputs with similar divergence from the ground truth.
- $\theta(x) < 1$: The deepfake model output is more misleading or deceptive than the hallucination model.

This θ metric provides an interpretable signal for measuring overfitting-induced drift. A high $\bar{\theta}$ across samples may reveal systematic hallucination bias in the overfitted model.

Applications of θ -Based Analysis

The θ distribution and its aggregate behavior enable:

- Threshold selection for hallucination detection
- Identification of high-divergence outlier samples
- Input to downstream classifiers to distinguish hallucinations from deepfakes
- Insight into modality-specific sensitivity to hallucination bias

Thus, the θ metric serves as both a diagnostic tool and a quantifiable lens through which hallucination bias can be evaluated in generative systems.

V. RESULTS AND DISCUSSION

Classification Performance

Both the hallucination-prone and deepfake-generating models demonstrated strong performance in classifying real versus manipulated face images. The deepfake model, trained with regularization on tampered data, achieved an overall accuracy of 98.5% on the test set. Notably, the hallucination model—despite being overfitted—exhibited nearly identical performance. The confusion matrices for both models revealed balanced precision and recall across real and fake classes, indicating that overfitting did not significantly degrade surface-level classification capability.

Divergence Analysis Using θ

To assess reconstruction fidelity, we computed the hallucination–deepfake divergence metric θ for each input sample. The histogram of θ values, shown in Fig. 1, is highly right-skewed with a long tail. Most values are concentrated near zero, suggesting that hallucinated outputs frequently diverge more from the ground truth than those of the deepfake model. Outlier samples exhibit extremely high θ scores, further reinforcing the hypothesis that overfitted models generate less reliable and less faithful reconstructions—even when classification accuracy remains high.

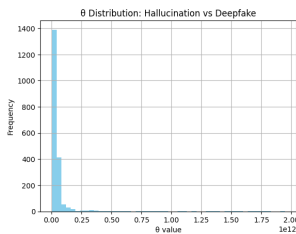


Figure 1. Histogram of θ values showing divergence distribution between hallucinated and deepfake outputs.

Latent Space Visualization

We visualized the latent embeddings of the deepfake model using t-SNE, as shown in Fig. 2. The plot reveals clear separation between real and fake data clusters, indicating that the model effectively learns discriminative latent representations. In contrast, the hallucination model (not shown) exhibited more dispersed and irregular clustering, suggesting that overfitting compromises the consistency and reliability of internal features. This highlights a critical insight: even if surface-level outputs appear accurate, internal representations can degrade under overfitting, making hallucination a hidden but measurable threat.

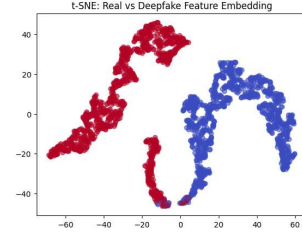


Figure 2. t-SNE visualization of latent embeddings from the deepfake model. Clear separation is observed between real and fake samples.

Statistical Validation

To statistically validate our findings, we conducted a Mann–Whitney U test comparing the reconstruction loss distributions of the hallucination and deepfake models. The test produced a U statistic of 4,000,000 with a p-value less than 0.00001, confirming that hallucinated outputs diverge significantly more from the ground truth than deepfake outputs. This strong statistical significance supports the use of θ as a robust and interpretable divergence metric for hallucination bias detection.

Summary of Findings

Together, these results confirm that overfitting in generative models—while not always detectable through standard accuracy metrics—can be effectively exposed through reconstruction-based analysis. The proposed θ metric, supported by visual and statistical evidence, provides a quantitative means of identifying hallucination bias and offers a new lens through which generative model reliability can be assessed.

VI. EMPIRICAL VALIDATION OF HYPOTHESIS

This research was grounded in the hypothesis that overfitted generative models are more prone to producing hallucinated outputs that deviate significantly from the ground truth—potentially even more than those produced by deepfake-generating models trained on manipulated data. While classification metrics alone showed high accuracy for both models, they were insufficient to expose deeper model behavior differences.

To validate this hypothesis, we introduced the divergence parameter θ , defined as the ratio of reconstruction errors

between the hallucination-prone and deepfake models. The distribution of θ values revealed that hallucination-induced outputs consistently exhibited greater divergence from original inputs, as evidenced by the right-skewed histogram (Fig. 1).

Furthermore, statistical analysis using the Mann–Whitney U test confirmed the significance of this divergence. The resulting U statistic of 4,000,000 and p-value less than 0.00001 demonstrate that the hallucination model’s outputs are not only qualitatively different but also statistically more divergent than those of the deepfake model.

This empirical evidence—spanning visual, mathematical, and statistical validation—strongly supports our central hypothesis. It shows that overfitting introduces a quantifiable hallucination bias, which can be systematically measured using the proposed θ metric. This insight contributes a novel perspective to evaluating generative models beyond surface-level performance metrics.

VII. CONCLUSION AND FUTURE WORK

In this study, we proposed and validated a novel divergence metric, θ , to quantify hallucination bias in generative models. By training two autoencoder-based systems—one overfitted on real data (hallucination-prone) and the other well-regularized on manipulated data (deepfake generator)—we demonstrated that surface-level classification accuracy is not always a reliable indicator of model integrity.

The introduction of the θ metric allowed us to compare the fidelity of model outputs against original inputs in a systematic manner. Our findings revealed that hallucinated outputs consistently diverge more from the ground truth than deepfake outputs. This was visually confirmed through the distribution of θ values and t-SNE projections, and statistically validated using the Mann–Whitney U test ($p < 0.00001$). Together, these results support the central hypothesis that overfitting contributes to measurable and often hidden hallucination bias in AI-generated media.

This work offers a fresh lens through which generative model outputs can be audited, especially in high-risk domains such as misinformation, identity manipulation, and synthetic media verification.

Future Work

Building upon these results, several directions for future research emerge:

- **Multimodal Hallucination Detection:** Extending the θ metric to audio, text, or video streams to detect cross-modal inconsistencies.
- **Model-Agnostic Evaluation:** Applying θ to other generative architectures (e.g., GANs, transformers) to assess generality.

- **Robust Training Techniques:** Using θ as a diagnostic tool during training to monitor and mitigate overfitting-induced hallucination.
- **Ethical AI Auditing:** Incorporating divergence metrics into digital forensic pipelines for detecting malicious or unreliable synthetic content.

Ultimately, this research highlights the importance of moving beyond accuracy to understand the subtle behaviors of generative models and reinforces the need for interpretability and accountability in synthetic media systems.

REFERENCES

- Cardenuto, J. P., et al. (2023). The Age of Synthetic Realities: Challenges and Opportunities. **APSIPA Transactions on Signal and Information Processing**, 12(1). <https://doi.org/10.1561/116.00000138>
- Carpenter, P. (n.d.). **FAIK: A Practical Guide to Living in a World of Deepfakes, Disinformation, and AI-Generated Deceptions.**
- Salvagno, M., Taccone, F. S., Gerli, A. G. (2023). Artificial intelligence hallucinations. **Critical Care**, 27(1), 180. <https://doi.org/10.1186/s13054-023-04473-y>
- Pataranutaporn, P., et al. (2024). Synthetic Human Memories: AI-Edited Images and Videos Can Implant False Memories. **arXiv preprint**. <https://doi.org/10.48550/arXiv.2409.08895>
- Shree, S., Arya, R., Roy, S. K. (2024). Investigating the Evolving Landscape of Deepfake Technology. **IRJAEH**, 2(5), 1489–1511. <https://doi.org/10.47392/IRJAEH.2024.0206>
- Liu, Z., Dou, G., Tan, Z., Tian, Y., Jiang, M. (2024). Machine Unlearning in Generative AI: A Survey. **arXiv preprint**. <https://doi.org/10.48550/arXiv.2407.20516>
- Ulhaq, A. (2021). On Deep Research Problems in Deep Learning. **engrXiv**. <https://doi.org/10.31224/osf.io/r78ds>
- FaceForensics++ Team. (2024). *FaceForensics++*. Kaggle. Retrieved from <https://www.kaggle.com/datasets/hungle3401/faceforensics>
- Gao, Y., Wang, J., Lin, Z., Sang, J. (2024). AIGCs Confuse AI Too. In **Proceedings of the 32nd ACM International Conference on Multimedia**, 9010–9018. <https://doi.org/10.1145/3664647.3681467>
- Lomnitz, M., et al. (2020). Multimodal Approach for DeepFake Detection. In **2020 IEEE AIPR Workshop**, 1–9. <https://doi.org/10.1109/AIPR50011.2020.9425192>